Forecasting the Outcome of an IPL Match using Machine Learning Techniques

[1]Srinath Kamalakumar, [2]Darshan Venugopal, [3*]E. Elakiya

[1, 2]School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, Tamil Nadu, India.
[3*]Assistant Professor, School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, Tamil Nadu, India.

[3*]elakiya.e@vit.ac.in

Abstract

Data Science is evolving and so companies from every field are now utilizing new technologies to improve their business. To enhance these benefits, analyze the data clearly and precisely. A growing field is machine learning which uses available data to predict future states and guide decisions based on what a model predicts. Cricket is one of the most popular sports in the world, watched across 104 countries, while the Indian Premier League (IPL) is the biggest cricket tournament in the world. This is a game of luck and the lovers and fans of the sports always want to know the result of the games. This study performs an extensive evaluation of diverse dataset features in predicting Cricket match results. To predict the outcomes of matches, we utilize carefully chosen features for the training of the machine learning (ML) models. Various machine learning models like Random Forest, Logistic Regression, SVM, KNN and Decision Tree are applied on different sized test and training datasets. We have tried several classification algorithms to improve prediction accuracy. Analytics are derived from these findings that are useful for cricket boards as well in estimating team strength and analytics of cricket. Moreover, for every player to identify their weakness and improve their game.

Keywords: Cricket, Indian Premier League, Match Prediction, Classification, Machine Learning.

1. Introduction

Machine learning, a crucial component of artificial intelligence, enables computers to learn and adapt independently, predict, and decide without explicit programming. Its importance in today's world is immense, supporting various groundbreaking technologies like recommendation systems, virtual assistants, autonomous vehicles, and healthcare diagnostics. By analyzing vast datasets, machine learning algorithms detect intricate patterns, offering valuable insights that enhance processes and decision-making in diverse industries. Recent advancements in computer science, the availability of extensive data, and theoretical progress have sparked a renewed interest in machine learning methods in the 21st century.

Cricket, originating in the 15th century and gaining prominence in England, holds the position of the second most popular sport worldwide after football. With its expanding global fan base, cricket has the potential to rival football in popularity. In contemporary India, cricket has transcended its status as just a game, evolving into a cultural phenomenon, almost resembling a religion. The introduction of the Twenty20 format in 2006 revolutionized the sport, with India securing victory in the inaugural world championship in 2007. This faster-paced format, featuring 20 overs per team and completing matches in under three hours, gained significant

Electronic copy available at: https://ssrn.com/abstract=5091637

traction, especially through the Indian Premier League (IPL) initiated in 2008. With its thrilling matches, star-studded line-ups, and passionate fan following, the IPL has transformed cricket, providing a global stage for emerging talents. IPL viewers often engage in predicting match outcomes using available facts, statistics, and records. Consequently, there's a substantial market for algorithms predicting scores and winning teams. However, the complexity of numerous influencing factors poses challenges for precise forecasting, and the accuracy of predictions relies heavily on the volume and quality of data employed in the process.

2. Related Works

Match prediction has become a very interesting aspect for Sports analysts over the evolution of cricket. Now, for the prediction of a cricket match outcome, several techniques have been used, e.g., Random Forest Classifier, SVM, Logistic Regression, KNN, Decision Tree, etc. [11] explore various machine learning algorithms, explain how they work and how to implement them. In their research, they create a model based on a training dataset, and with this model, they can only predict by applying it. This methodology is composed of five sub-modules: dataset loading, pre-processing, feature selection, classification with several algorithms and algorithm comparison. The Data was imported using pandas and preprocessing was done by Amit Nagarale, Mandar Pardakhe, Nikita Wavre, Shraddha Dhole and Nikhil Dhonge [7]. They took care of nulls by filling them with mean. They utilized math equations to compute entropy and derived information gains. Using Machine Learning with Random Forest Classifier, SVM, Logistic Regression, KNN and Decision Tree, they predicted the IPL match results. It was the work of Leo Gertrude David, Faizan Rasheed, and Daniel Mago Vistro [10] who found out that Decision Tree is the most accurate predictor for the IPL match. They used a more potent estimation and averaging strategy by running different decision tree classifiers on sub-samples of the entire dataset. They concentrated on preprocessing the IPL T20 match winner dataset to make sure it was in a consistent format including converting variables into numerical format and removing missing values. Identifying Optimal Features: The Approach Ideal Linear Classifier Kaluarachchi et al. Introduction: [5] created predictions for match results considering home ground advantage, match hour, match type, Naïve Bayes classifier, toss win and first bat. Player performance was handled by Passi et al. [6] prediction in which treating runs and wickets are considered classification problems using ML algorithms. They observed that Random Forest outperformed other algorithms in this specific task.

Aman Sahu, Devang Kaushik, and A. Meena Priyadharsini [4] proposed a system addressing the challenges of maintaining individual player records by allowing users to upload historical data via machine learning tools like Pycharm, Jupiter, or Google Colab. They performed data analytics on this system to extract meaningful insights.

Kiran Gawande, Prof. Sumit Harale and Simran Pakhare [12] in order to counteract the constraints posed by the limited IPL dataset, extracted critical features based on domain expertise, utilizing data mining methods. Understanding the significance of this restricted dataset, multiple tiers of features were crafted to ensure the resulting model wasn't underfit. This approach aimed to incorporate various levels of detail and nuances, enriching the model's capacity to capture the intricacies of IPL match outcomes despite the dataset's size limitations. Harshal Patel, Divy Tolani, Baibhav Kumar, Aryan Khetan, and Srikantaiah K C [3] watched an IPL game outcome based on stadium and toss decisions, employing machine learning techniques like SVM, Random Forest, and Logistic Regression. Their research highlighted

Electronic copy available at: https://ssrn.com/abstract=5091637

Random Forest achieving the highest accuracy of 88.10% and implementing various cross-validation techniques for the classification. Rameshwari Lokhande [13] devised a strategy to increase user participation in the Fantasy Cricket game and to enhance overall engagement with the Premier League. Through the predictive model, users are motivated to make predictions for every game, fostering increased viewership as they tune in to validate their predictions during match viewing. Dr. Chandrani Singh, Dr. Ramesh D Jadhav, Dr. Bharti P Jagdale, Dr. Sunil Khilari [14]. With machine learning methods like Ridge Regression, Linear Regression, and Random Forest, the study aims to predict IPL match winners based on variables like toss results, venue, and player behaviors.

K Rushikesh Reddy, Chandrakanth V, A. Prem Sai, and Shiva Sumanth Reddy [2] conducted a team-wise analysis to uncover strengths, weaknesses, and game developments during different stages of the match. They utilized IBM Watson for their analysis to aid team management in making informed decisions. Mei-Ling Huang and Yun-Zhi Li [15]. This research pioneers the use of a 1DCNN model in forecasting MLB match results, uniquely leveraging its ability to automatically extract features. It establishes two datasets for comparison: one considering only Starting Pitcher (SP) data and another encompassing all pitchers' data. Results indicate that employing data from all pitchers leads to superior prediction accuracy. Utilizing Baseball-Reference.com's data on hitters and pitchers, this study's three proposed models surpassed related studies in predicting MLB match outcomes, demonstrating higher accuracy. Fatima Rodrigues and Angelo Pinto [1] predicted football match outcomes using a machine learning algorithm. They incorporated team performance metrics and player statistics from 2013/14 to 2018/19. Their model initially tested all variables and narrowed down to 18 key variables for further testing and combination exploration, achieving a 65.26% correct prediction rate. In Chen's study [16], For the purpose of selecting features, logistic regression was used, and the significance of each independent variable was determined by sensitivity analysis. For the analysis, variables with a relative relevance of greater than 80% were selected. The literature on ML trends and techniques for predictive analysis is reviewed in this paper. In order to provide a review that reflects the current state of research, we adopted Patrick Loola Bokonda, Khadija Ouazzani-Touhami, Nissrine Souissi [17]. This research conducts a comprehensive literature review on the prevalent trends and methodologies in machine learning (ML) utilized for predictive analysis. To ensure an up-to-date overview of the research landscape, the study focused on identifying the primary disciplines that commonly employ ML methods for predictive analysis purposes. Slamet Wiyono, Dega Surono Wibowo, M. Fikri Hidayatullah and Dairoh [18] used various machine learning algorithms like SVM, KNN and Decision Tree to evaluate the student performance. L. Eswar Kumar, V. Yagnesh, U. Likith, M. Suresh [19] conducted a study employing Classification Algorithms such as SVM, Random Forest, Decision Tree, logistic regression, , k-nearest Neighbor and XGBoost to predict the final outcome of matches. Their findings revealed that the decision tree algorithm exhibited the highest accuracy among the tested models. Mayank Agarwal, Prof. Dr. Archana Kumar [20]. Machine learning methods are employed to forecast the outcome of the match using the IPL dataset with various methods like Regression, Linear Regression by considering factors like ground history, team balance, current situation, run rate, overs remaining and so on.

Electronic copy available at: https://ssrn.com/abstract=5091637

3. Materials and methods

3.1 Dataset Description

The IPL dataset comprises match statistics and winner details, encompassing team participation, victorious teams, match venues, victory margins in wickets and runs, toss decisions, implementation of the Duckworth-Lewis-Stern (DLS) method, and umpire names. Feature engineering on the IPL dataset yielded key insights into match dynamics like the number of times when teams won the toss or teams who won the match after winning the toss.

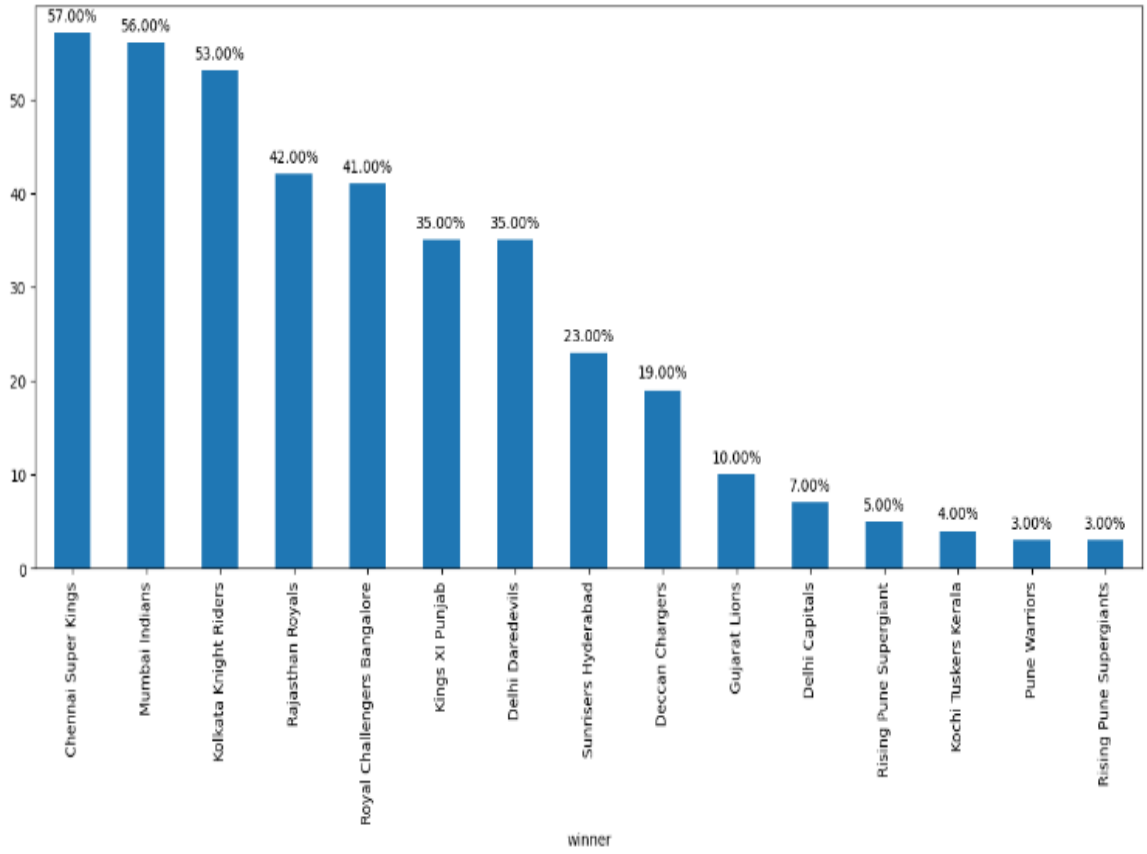


Figure 1. Teams that won after winning the toss.

Figure 1. show the Teams that won after winning the toss.

As we can see when we analyze the dataset for the number of wins for the teams that have won the toss the highest is that for the Mumbai Indians, which is over 120. Sunrisers Hyderabad, on the other hand, are also running to keep up with them in a close contest with Delhi Capitals battling in their ranks. Kolkata Knight Riders, Chennai Super Kings, Royal Challengers Bangalore are also doing very well in the same race. Punjab Kings possess a decent win's tally, but both LSG and GT are at the bottom, vindicating their newer nature of playing in IPL. The Kochi Tuskers Kerala at the bottom of the list, is a testament to their lack of impact in the IPL. This is based on analyzing the number of matches IPL teams won while batting first. Mumbai Indians are on the top followed closely by Chennai Super Kings and Kolkata Knight Riders with all other teams following a close but similar trend. This emphasizes how strong these teams are when they have a target for the opponent.

Electronic copy available at: https://ssrn.com/abstract=5091637

3.2 Architecture Design

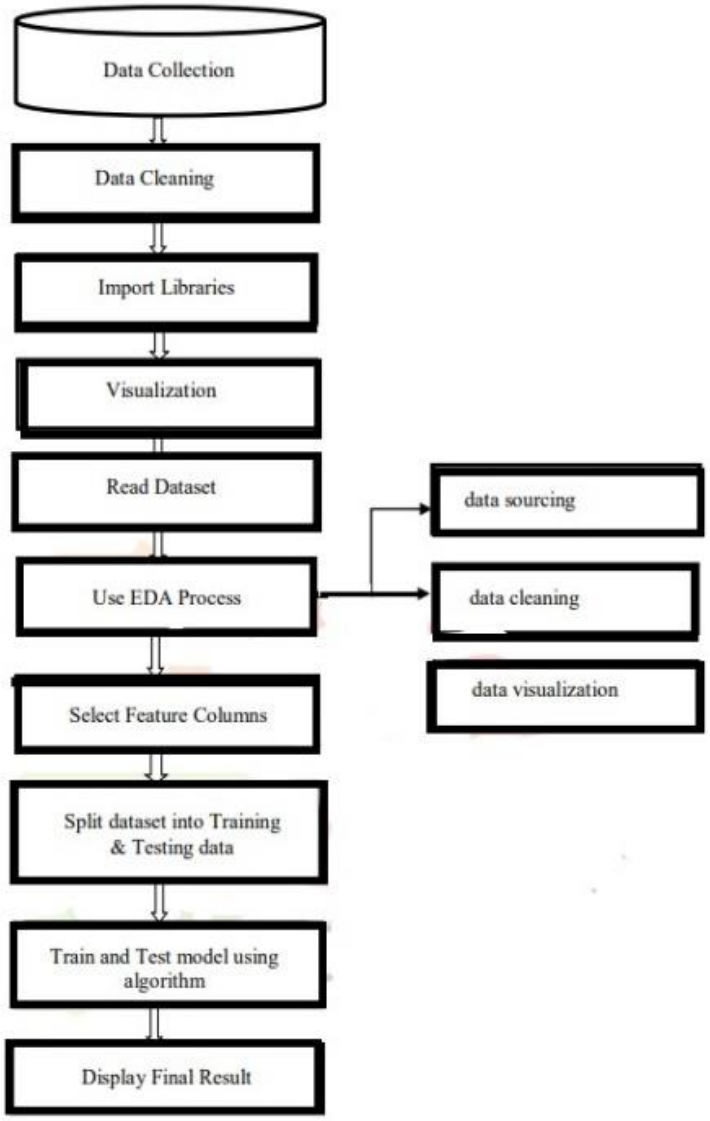


Fig: System Architecture

Figure 2. System Architecture

During the initial step, a data set recording historical match results from 2008 to 2016 was compiled. After data acquisition, relevant Python libraries such as Pandas, Matplotlib and Seaborn were imported for analysis of this dataset. The key focus was to get a better understanding of data insights by using deep data visualization methods, using Matplotlib and Seaborn. After loading the dataset and showing it in tabular form, a detailed Exploratory Data Analysis (EDA) was performed to discover important patterns and fix any data issues. The EDA Flow data sourcing and loading procedures were carried out to derive comprehensive data characteristics, such as entry counts, column details, null, and data types per column. Then comes the profiling and cleaning of data: Eliminating null values, avoiding the use of unimportant columns, among others to increase prediction quality. This preparation stage ended with the plotting of the final dataset. The next step consisted of choosing which feature columns

Electronic copy available at: https://ssrn.com/abstract=5091637

to include, and I decided to encode the values of "team1," "team2," "city," "toss winner," "toss decision," and "venue" as independent variables, while choosing "winner" as the dependent variable. To facilitate training and evaluation of machine learning models, the dataset was split into a distinct training and testing split, with 20% of data held out for testing and 80% for training. Lastly, a machine learning algorithm was applied, which aimed at predictive insights. (Figure 2).

4. Models Description

4.1 Random Forest

Random Forest classification entails multiple decision trees created upon various dataset subsets. It's a technique used for regression as well as classification. In this method, each tree in the Random Forest processes an input vector by voting on the class label or target variable for a new instance. The final classification is determined by the node receiving the highest number of votes among the trees. To enhance prediction accuracy and combat overfitting, Random Forest employs an estimation and averaging approach on dataset sub-samples equal in size to the original input [10] [11]. Each feature importance in a decision tree calculation is done by the formula.

$$\mathbf{fi_i} = \frac{\sum_{\mathbf{j:node\ j\ splits\ on\ feature\ i}} \mathbf{ni_j}}{\sum_{\mathbf{k\in all\ nodes}} \mathbf{ni_k}} \quad (1)$$

4.2 Logistic Regression

Logistic regression is a statistical/machine learning model used for binary classification. Its main goal is to provide prediction on one of the two possible outcomes often referred into 0 (representing a negative class) or 1 (positive class). This technique is widely used in many areas of machine learning and is commonly used for spam detection, medical diagnosis, etc. After training the model with labeled data, the model can make predictions on new, unseen data points based on the patterns it has learned. The model maps input features to log-odds and passes it through the sigmoid, producing a probability for class positive. If this probability is more than a threshold (generally 0.5), the model predicts class 1; otherwise class 0. It performs appropriately when the association between the features and the log-odds of the response variable is approximately linear.

4.3 KNN

The k-nearest neighbors, leverages the power of nearest neighbors, employing supervised learning for robust classification that operates without explicit parameters and relies on proximity to categorize individual data points. While usable for both regression and classification, it's primarily employed as a classification tool, assuming that similar data points are close to each other. The 'k' value in KNN determines how many neighboring points are considered when classifying a query point. The choice of 'k' depends largely on the dataset, as data with more outliers or noise tend to benefit from higher 'k' values for improved performance [9]. Euclidean distance for the distance between two points

$$\text{Euclidean Distance} = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} \quad (2)$$

Electronic copy available at: https://ssrn.com/abstract=5091637

4.4 SVM

Support Vector Machine (SVM) is one of the most widely used classifiers in Ada Boosting for making prediction as it can be used to make many types of prediction such as image recognition, medical diagnosis, constellate etc. The SVM classifier finds an optimal hyperplane on the training data, which is then used to classify new examples. The space is divided into two sections with this hyper-plane acting as the separator with each class falling alongside the hyper-plane. Support Vector Machine is measured for its performance through its Regularization parameters that affect directly on the optimization process. SVM is a supervised machine learning algorithm, which means that it requires pre-defined output classes for training. It is best to use to create a perfect hyperplane with training data that divides the space into classes to successfully classify the other instances. [10]

4.5 Decision Tree

It is a visual representation of a decision-making process, where a tree-like structure is created to model various decisions and their potential consequences the heart of a decision tree's accuracy lies in its strategic branching strategy. This branching, however, hinges on distinct criteria for classification and regression trees. Both types leverage diverse algorithms to discern whether a node should be partitioned into multiple sub-nodes. This partitioning aims to maximize homogeneity, essentially boosting purity with respect to the target variable. This process involves the tree splitting nodes across all accessible variables and selecting the split that generates the most homogeneous sub-nodes. [8]

5. Results

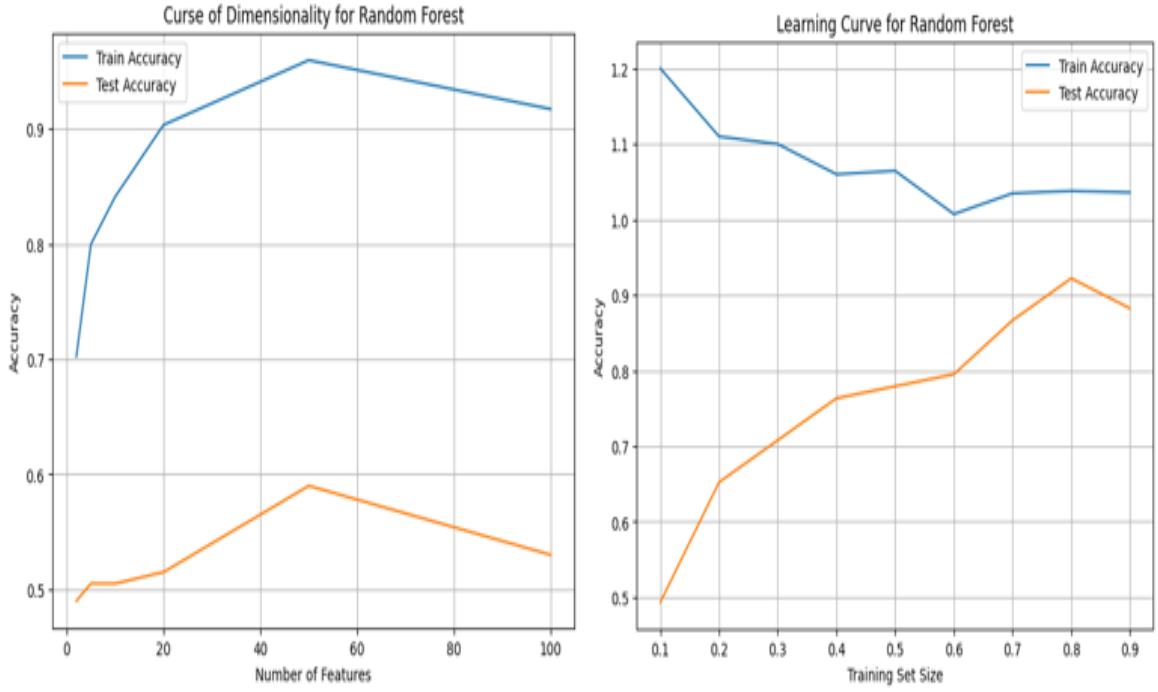


Figure 3. Accuracy and Loss Graph for Random Forest

Electronic copy available at: https://ssrn.com/abstract=5091637

The training and testing accuracy and loss of various models are shown in Figure 3 to 6. The confusion matrix of all the models is represented in the Figure 7 to 11. Table 1 represent the comparison of accuracy and error and shown in Figure 12.

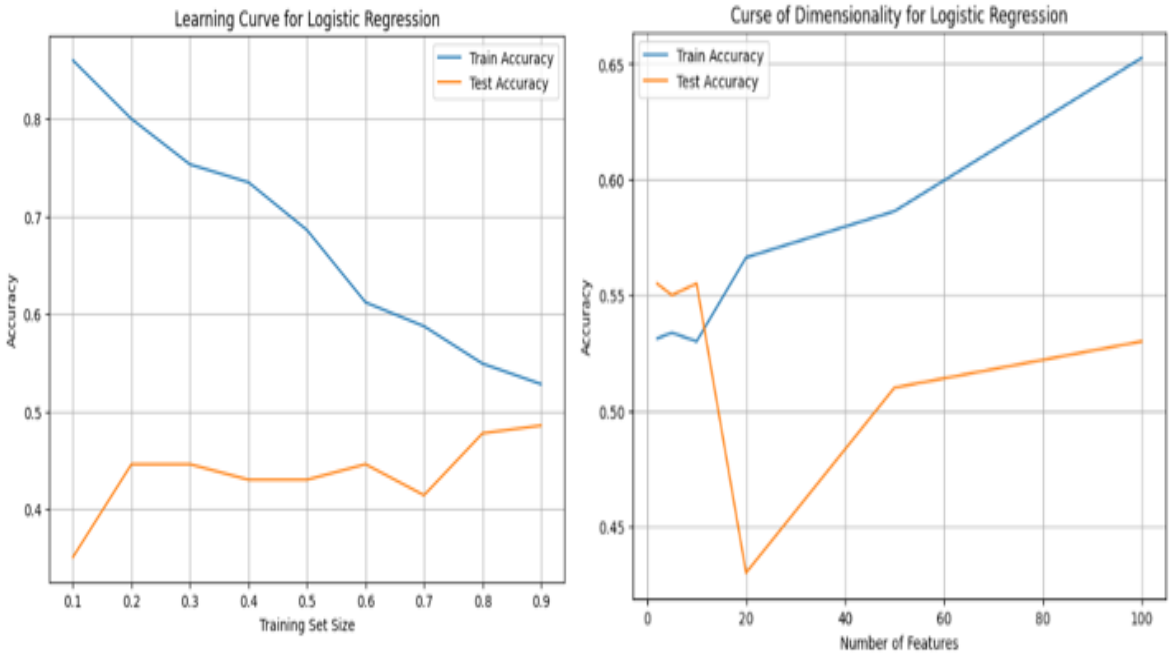


Figure 4. Accuracy and Loss Graph for Logistic regression

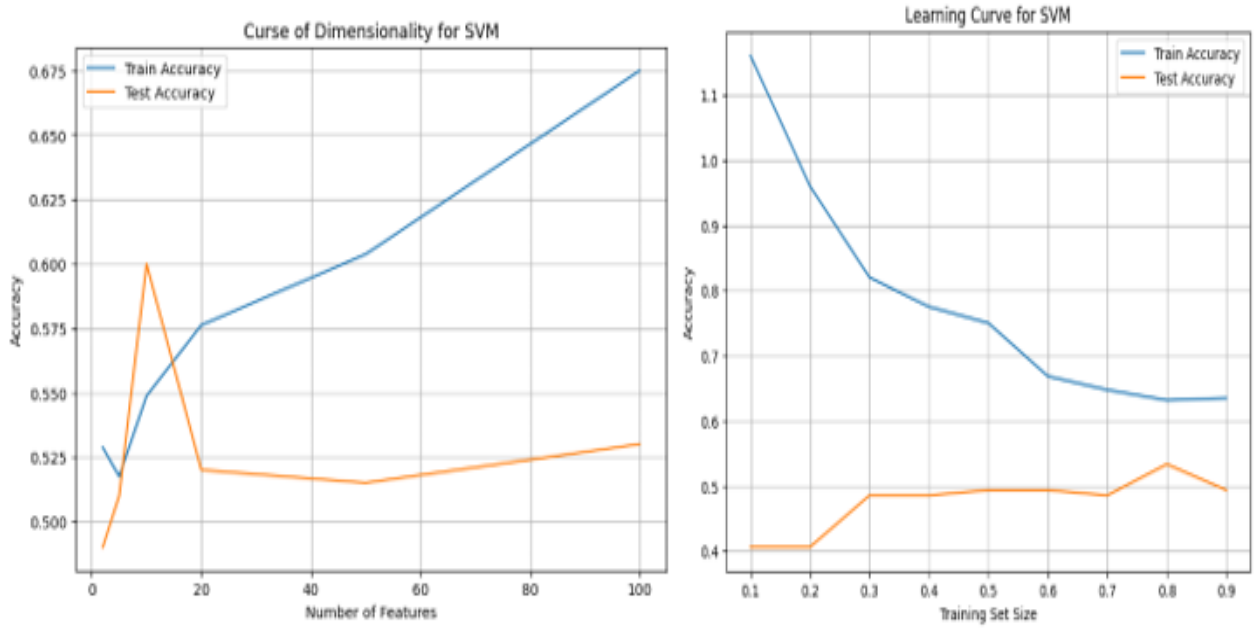


Figure 5. Accuracy and Loss Graph for SVM

Electronic copy available at: https://ssrn.com/abstract=5091637

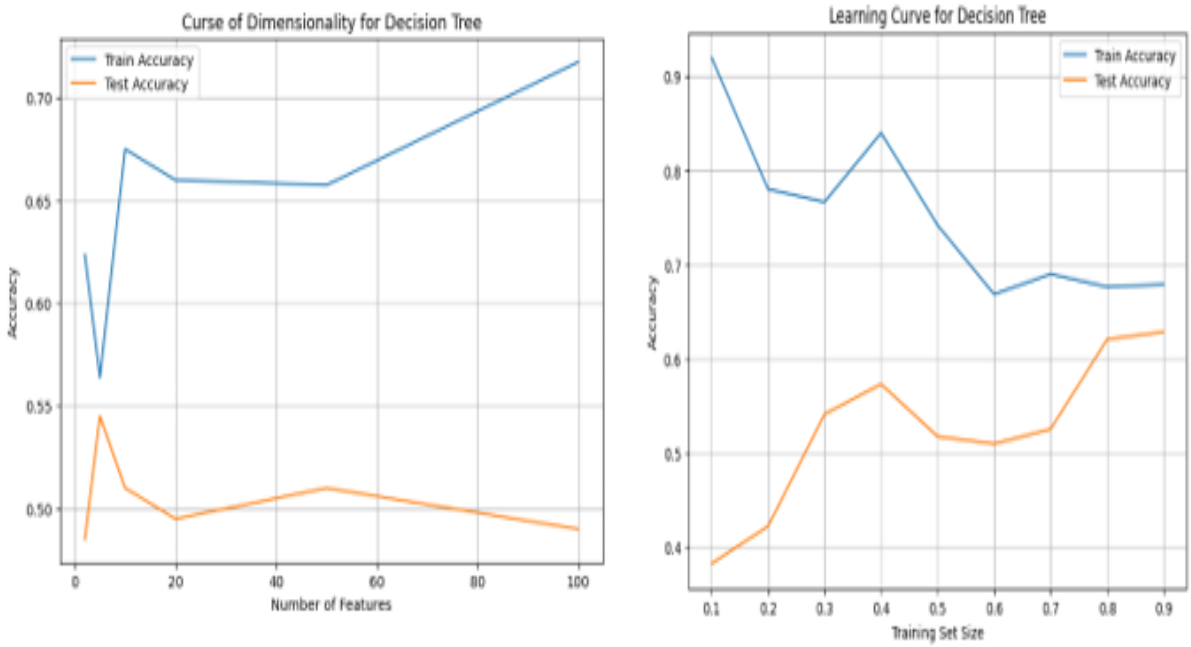


Figure 6. Accuracy and Loss Graph for Decision Tree

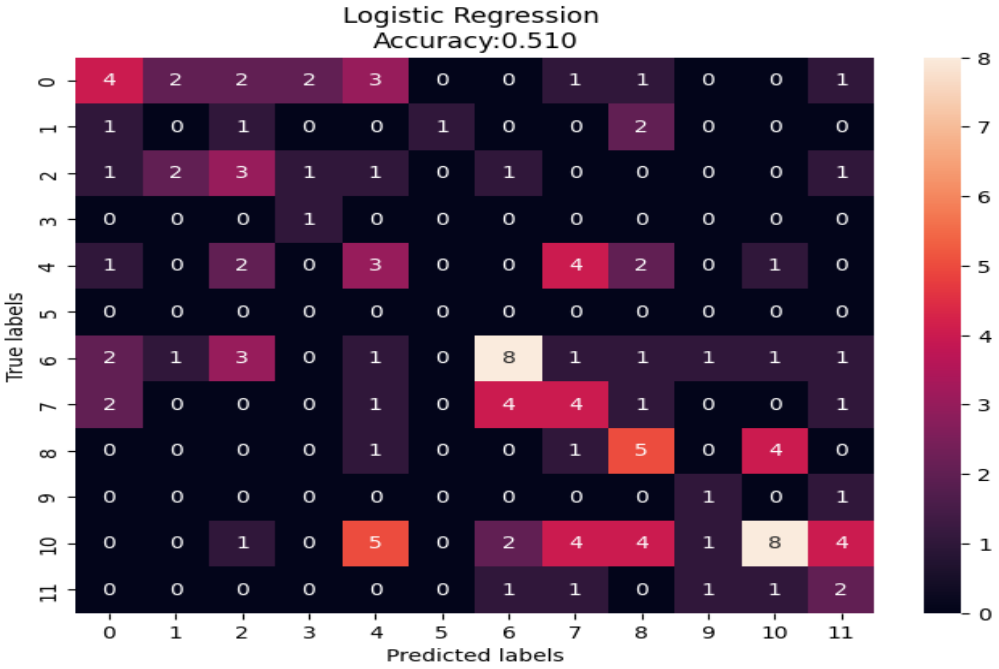


Figure 7. Confusion Matrix for Logistic Regression

Electronic copy available at: https://ssrn.com/abstract=5091637

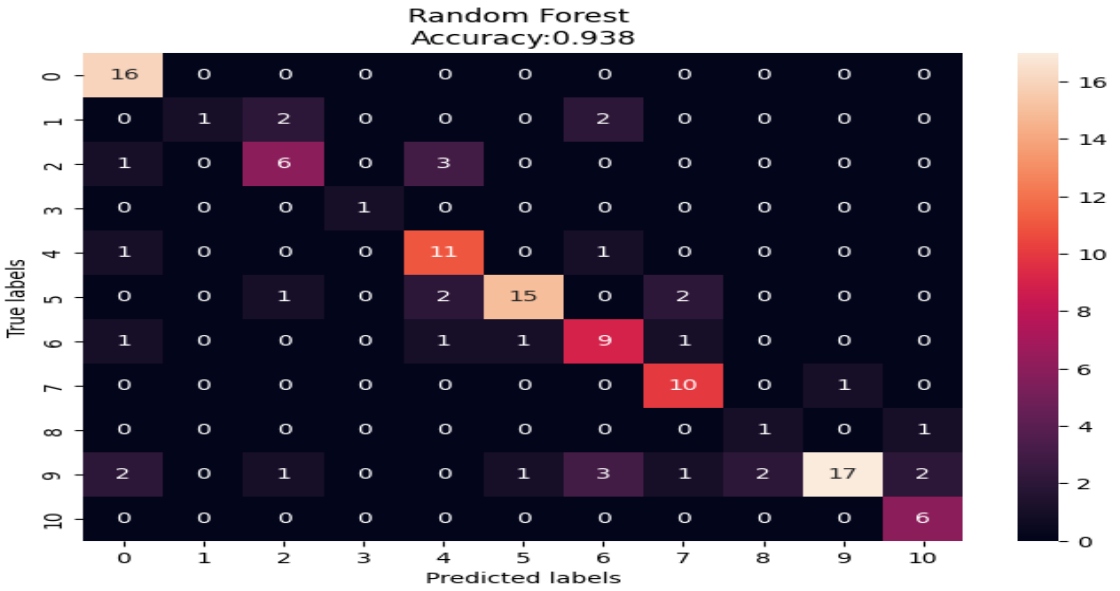


Figure 8. Confusion Matrix for Random Forest

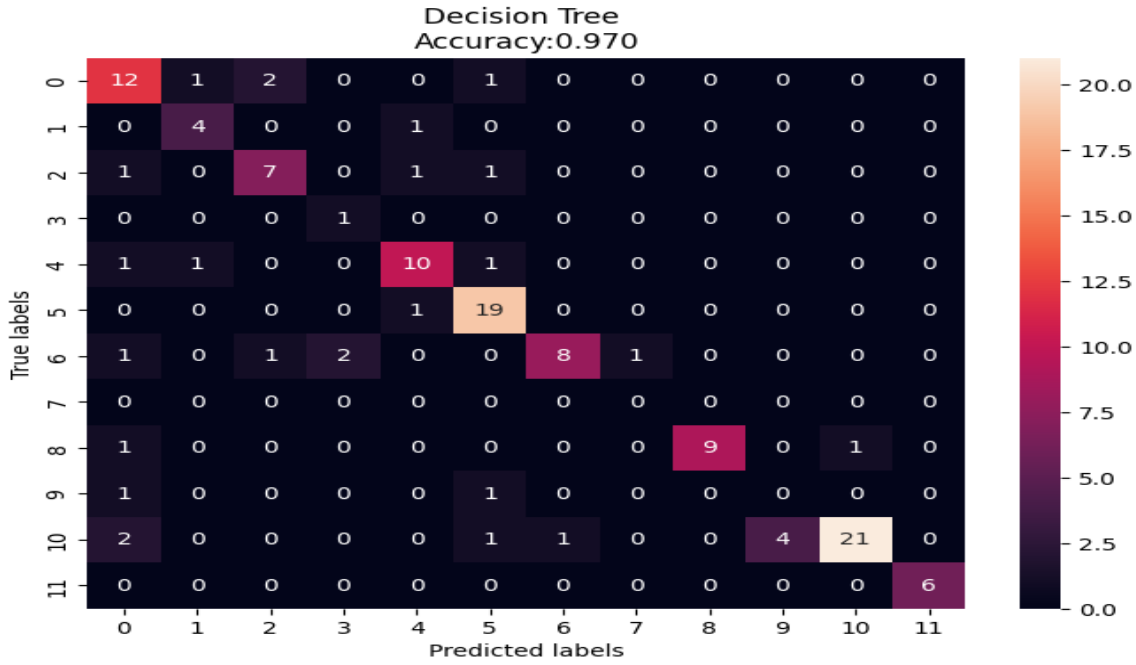


Figure 9. Confusion Matrix for Decision Tree

Electronic copy available at: https://ssrn.com/abstract=5091637

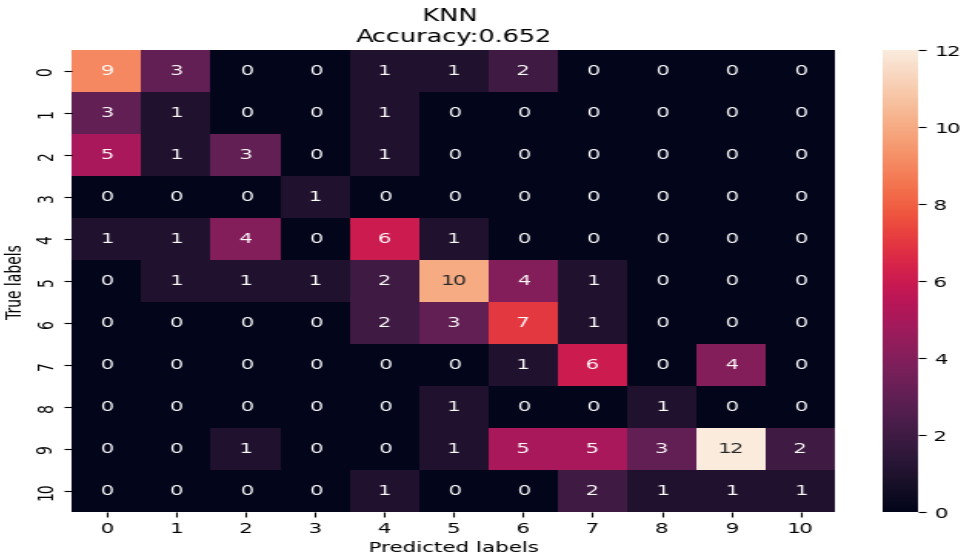


Figure 10. Confusion Matrix for KNN

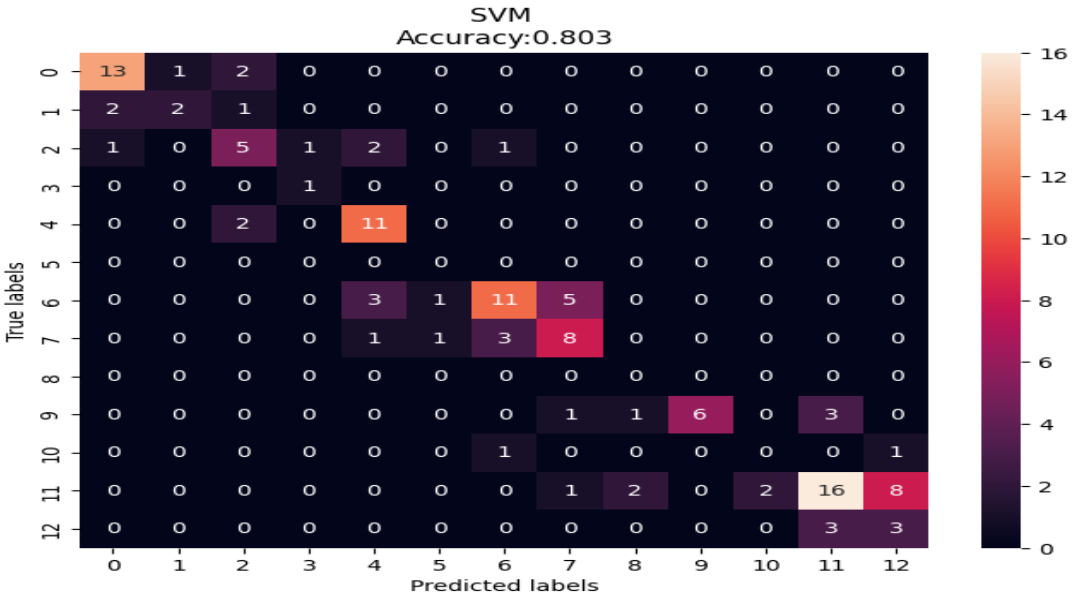


Figure 11. Confusion Matrix for SVM

Electronic copy available at: https://ssrn.com/abstract=5091637

Table 1. Comparison of used models Accuracy and Error

Model	**Accuracy** (%)	**Error** (%)
Logistic Regression	51	49
Decision Tree	97	3
Random forest	94	6
KNN	66	34
SVM	80	20

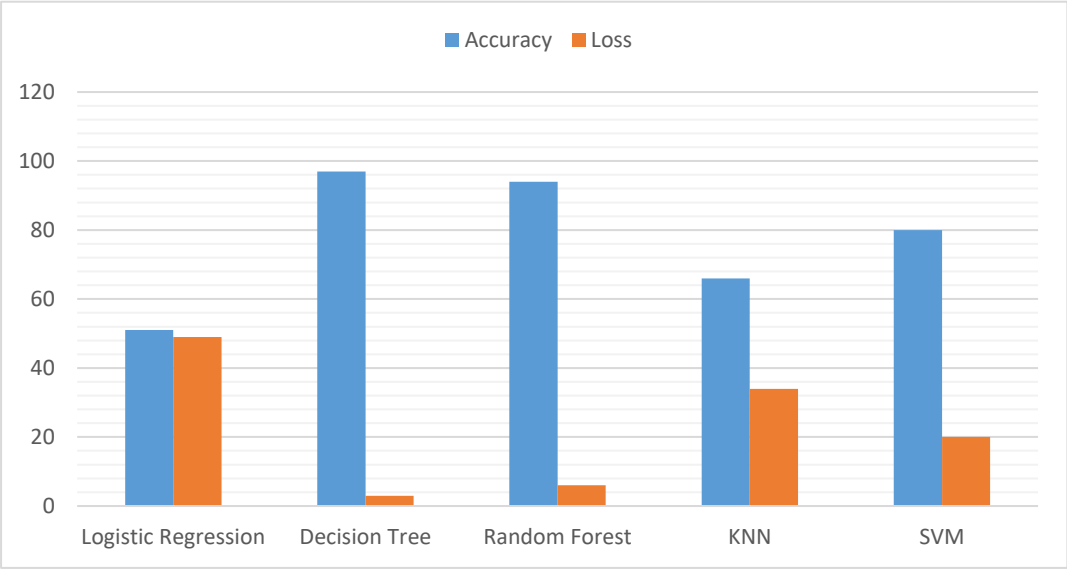


Figure 12. Accuracy Comparison of used Models

6. Conclusion

The multilayered nature of cricket makes it difficult to predict the winner in this game, hence, predicting the winner becomes one of the biggest challenges. Nonetheless, the rise of machine learning provides a new opportunity to ease and automate this process. We discovered many factors that influence IPL match outcomes in this study. The outcome of an IPL game is determined by a thousand factors apart from just playing fucking cricket. The players in this cricketing story are not limited to the two teams playing against each other but extend to the venue itself in which the contest takes place (the stadium), the unique atmosphere of the city and even the result of the coin toss. What turns it into competition is the constant watch of the umpire, so that chance comes to play a little more on the field. This complex interaction of talent, tactics, and luck makes every IPL match an enthralling event. Various machine learning methods are used for predicting the outcome of a match such as: Decision trees, Random forest, K-Nearest neighbors (KNN), Support vector machine (SVM) and Logistic regression.

Electronic copy available at: https://ssrn.com/abstract=5091637

References

1. Fatima Rodrigues, Angelo Pinto, 'Prediction of football match results with Machine Learning', International Conference on Industry Sciences and Computer Science Innovation, Procedia Computer Science 204 (2022) 463-470.
2. K Rushikesh Reddy, Chandrakanth V, A. Prem Sai, Shiva Sumanth Reddy, 'Super Predictor of Indian Premier League (IPL) using Various ML techniques with help of IBM Cloud', IJARIIE-ISSN(O)-2395-4396, Vol-8 Issue-2,2022
3. Srikantaiah K C , Aryan Khetan , Baibhav Kumar , Divy Tolani , Harshal Patel, 'Prediction of IPL Match Outcome Using Machine Learning Techniques', 3rd International Conference on Integrated Intelligent Computing Communication & Security (ICIIC 2021).
4. Aman Sahu, Devang Kaushik, A. Meena Priyadharsini, 'Predictive Analysis of Cricket', Turkish Journal of Computer and Mathematics Education Vol.12 No.6 (2021), 5111-5124.URL: https://turcomat.org/index.php/turkbilmat/article/view/8762/6831
5. Amal Kaluarachchi, S. Varde Aparna, 'CricAI: A classification-based tool to predict the outcome in ODI cricket', 2010 Fifth International Conference on Information and Automation for Sustainability, 17-19 Dec. 2010
6. Kalpdrum Passi and Nirav Kumar Pandey, 'Predicting player's performance in One day International cricket matches Using Machine learning', Volume 8, Computer Science & Information Technology, 2018.
7. Nikhil Dhonge, Shraddha Dhole, Nikita Wavre, Mandar Pardakhe, Amit Nagarale, 'IPL cricket score and winning prediction using machine learning techniques', International Research Journal of Modernization in Engineering Technology and Science Volume:03/Issue:05/May-2021.
8. Prince Yadav, "Decision Tree in machine learning", https://towardsdatascience.com/decision-tree-in-machine-learning- e380942a4c96, Nov 14, 2018.
9. Elakiya, E., et al. "Text Feedback Classification using Machine Learning Techniques." 2023 2nd International Conference on Edge Computing and Applications (ICECAA). IEEE, 2023.
10. Daniel Mago Vistro, Faizan Rasheed, Leo Gertrude David, 2019.The Cricket Winner Prediction With Application Of Machine Learning And Data Analytics. International Journal Of Scientific & Technology Research Volume 8, Issue 09, September 2019.URL: https://www.kclas.ac.in/wp-content/uploads/2021/01/The-Cricket-Winner-Prediction-With-Application-Of-Machine-Learning-And-Data-Analytics-.pdf
11. G. Sudhamathy and G. Raja Meenakshi,2020. Prediction on IPL Data using Machine Learning Techniques in R Package. ICTACT Journal on Soft Computing, October 2020, Volume: 11, Issue: 01.Url: https://ictactjournals.in/paper/IJSC_Vol_11_Iss_1_Paper_2_2199_2204.pdf
12. Kiran Gawande, Prof. Sumit Harale, Simran Pakhare,2022. Predictive Analysis of an IPL Match using Machine Learning.IJCRT Journal, Volume 10, Issue :4 April 2022.URL: IJCRT2204215.pdf
13. Rameshwari Lokhande, 2018.Prediction of live cricket score and winning. IJRTD Journal Volume 5(4), (2018).
14. Singh, Chandrani and Dr. Bharti P Jagdale, and Khilari, Dr.Sunil and Mr.Ravi Mourya, Mr.Ravi Mourya and Jadhav, Dr Ramesh D, Prediction of IPL Match Performance Based on Batsman Category Using Machine Learning Algorithm (March 21, 2022). Available at SSRN: https://ssrn.com/abstract=4062937 or http://dx.doi.org/10.2139/ssrn.4062937

Electronic copy available at: https://ssrn.com/abstract=5091637

15. Huang, M L & Li, Yun-Zhi. (2021). Use of Machine Learning and Deep Learning to Predict the Outcomes of Major League Baseball Matches. Applied Sciences.
16. Chen, C.-W,2014. Construction of the Winner Predictive Model in Major League Baseball Games: Use of the Artificial Neural Networks. Sport. Exerc. Res. 2014, 16, 167–181.
17. Bokonda, Loola & Khadija, Ouazzani Touhami & Souissi, Nissrine,2020. Predictive analysis using machine learning: Review of trends and methods. URL:http://dx.doi.org/10.1109/ISAECT50560.2020.9523703 .
18. Slamet Wiyono, Dega Surono Wibowo, M. Fikri Hidayatullah and Dairoh,2020. Comparative Study of KNN, SVM and Decision Tree Algorithm for Student's Performance Prediction. IJCSAM Vol. 6, No. 2, August 2020.URL: 5091 (its.ac.id)
19. L. Eswar Kumar, V. Yagnesh, U.Likith, M. Suresh, 2022. Cricket (IPL) Prediction based on Previous Data. Volume 5- Issue 2, Paper 50 August 2022.
20. Mayank Agarwal, Prof. Dr. Archana Kumar, 2023. IPL First Innings Score Prediction Using Machine Learning Techniques. Volume 9, Issue 2, Jan-Feb-2023. URL: Journal of Scientific Research & Engineering Trends Volume 3, Issue 6, Nov.-2017, ISSN (Online): 2395-566X (ijsret.com)

Electronic copy available at: https://ssrn.com/abstract=5091637